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In humans, the ABO blood groups and Rhesus blood groups are under genetic control. The inheritance of the ABO blood groups is controlled by three alleles of a single gene, I^A , I^B and I^O . The alleles I^A and I^B are codominant, and the allele I^O is recessive to I^A and recessive to I^B .

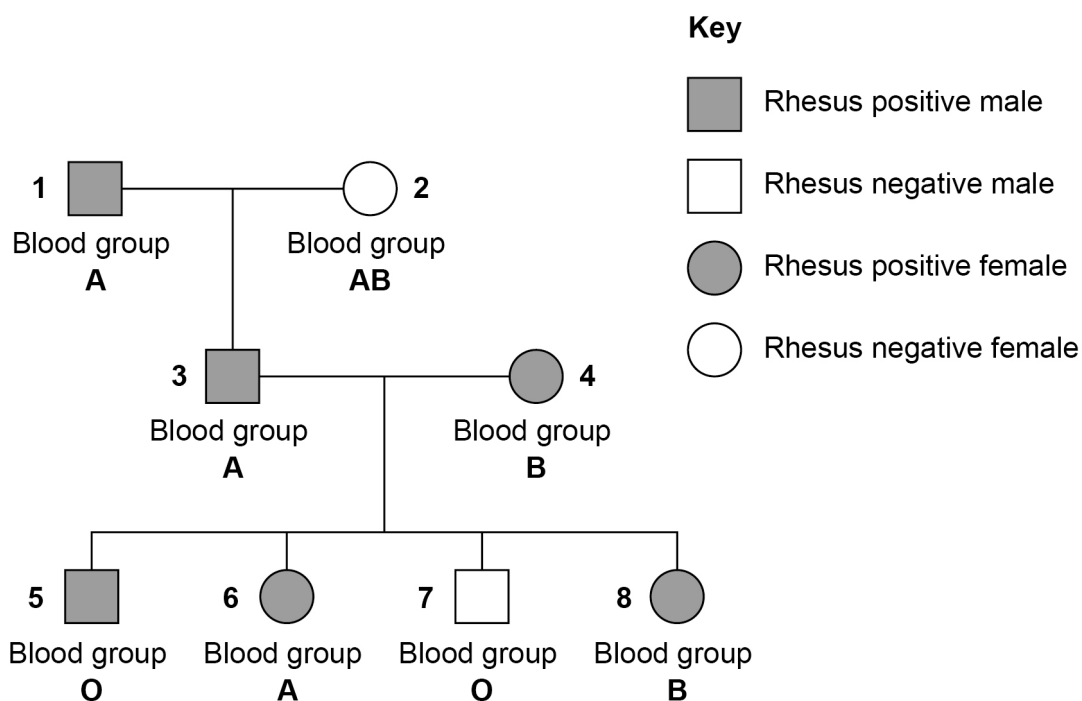
There are four ABO phenotypes, **A**, **B**, **AB** and **O**.

The gene for the Rhesus blood groups has two alleles. The allele for Rhesus positive, **R**, is dominant to the allele for Rhesus negative, **r**.

The genes for the ABO and Rhesus blood groups are **not** sex-linked and are **not** on the same chromosome.

Figure 3 shows the phenotypes in a family tree for the ABO and Rhesus blood groups.

Figure 3



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. 1

Give the genotypes of the ABO blood groups for individuals **1** and **2**.

Do **not** include the genotypes for the Rhesus blood groups in your answer.

[1 mark]

1 _____

2 _____

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Explain **one** piece of evidence from **Figure 3** that the allele for Rhesus positive is dominant.

[2 marks]

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Calculate the probability of individuals **1** and **2** producing a Rhesus positive son with blood group **A** (individual **3**). You can assume that individual **1** is heterozygous for the Rhesus blood group.

Show your working.

[2 marks]

Probability _____

Scientists determined the frequencies of the ABO alleles and ABO phenotypes in a large population. They then used a statistical test to determine if the frequencies of the four phenotypes differed significantly from the frequencies expected according to the Hardy–Weinberg equation.

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The frequencies of the I^A and I^O alleles were 0.15 and 0.65. What is the frequency of the I^B allele?

[1 mark]

Frequency of I^B allele _____

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Name the statistical test you should use to determine if the observed frequencies of the four phenotypes differed significantly from the frequencies expected according to the Hardy–Weinberg equation.

State how many degrees of freedom should apply.

[2 marks]

Statistical test _____

Number of degrees of freedom _____

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The scientists concluded that the observed frequencies of the four phenotypes differed significantly from the expected frequencies. Use your knowledge of the Hardy–Weinberg principle to suggest **two** reasons why.

[2 marks]

1 _____

2 _____
